

Assignment 3

- A. Design a 2nd-order feed-forward **continuous-time** sigma delta modulator for audio signals with a **3-bit** quantizer (mid-rise) and **thermometer** DAC.
- Noting that the audio BW extends from 20 Hz to 20 kHz, determine the minimum OSR for a peak SQNR of 100 dB. Use the corresponding maximum signal amplitude (MSA) while answering the following questions. [2 points]
 - What happens to the peak SQNR in the case of a **non-ideal** DAC whose unit-elements have a finite rise-time equal to 20% of a clock period and an ideal fall-time. [3 points]
 - With an **ideal** DAC, determine the input-referred thermal noise level that results in a peak SNR of 90 dB. Fix the SQNR at 100 dB. [1 point]
 - Scale the modulator so that the swing of both integrators is < 0.5 V. [2 points]
- B. Compare the modulator designed in (A) to a modulator with the same specifications but based on a **1-bit** quantizer. Comment on the differences. [2 points]
- **Submit your answers before the class on Thursday May 28th.**
 - **Instructions about submitting assignments are on Brightspace.**